

Review of improved Monte Carlo methods in uncertainty-based design optimization for aerospace vehicles



Xingzhi Hu^a, Xiaoqian Chen^{a,*}, Geoffrey T. Parks^b, Wen Yao^a

^a College of Aerospace Science and Engineering, National University of Defense Technology, Changsha 410073, China

^b Department of Engineering, University of Cambridge, England CB2 1PZ, United Kingdom

ARTICLE INFO

Article history:

Received 27 March 2016

Received in revised form

15 July 2016

Accepted 15 July 2016

Available online 27 July 2016

Keywords:

Uncertainty-based design optimization

Uncertainty analysis

Improved Monte Carlo

Sampling-based methods

Aerospace vehicle

ABSTRACT

Ever-increasing demands of uncertainty-based design, analysis, and optimization in aerospace vehicles motivate the development of Monte Carlo methods with wide adaptability and high accuracy. This paper presents a comprehensive review of typical improved Monte Carlo methods and summarizes their characteristics to aid the uncertainty-based multidisciplinary design optimization (UMDO). Among them, Bayesian inference aims to tackle the problems with the availability of prior information like measurement data. Importance sampling (IS) settles the inconvenient sampling and difficult propagation through the incorporation of an intermediate importance distribution or sequential distributions. Optimized Latin hypercube sampling (OLHS) is a stratified sampling approach to achieving better space-filling and non-collapsing characteristics. Meta-modeling approximation based on Monte Carlo saves the computational cost by using cheap meta-models for the output response. All the reviewed methods are illustrated by corresponding aerospace applications, which are compared to show their techniques and usefulness in UMDO, thus providing a beneficial reference for future theoretical and applied research.

© 2016 Elsevier Ltd. All rights reserved.

Contents

1. Introduction	20
2. General Monte Carlo based methodology	21
3. Improved Monte Carlo approaches	21
3.1. Bayesian inference	22
3.2. Importance sampling	22
3.3. Latin hypercube sampling	22
3.4. Meta-modeling approximation based MCS	23
4. Aerospace vehicle applications	23
4.1. Aerospace gas turbine engine prognostics	23
4.2. Reliability analysis of aircraft inside flap	24
4.3. RBMDO of aerospace launch vehicles	24
4.4. Uncertainty quantification in satellite system design	25
5. Conclusions	25
Acknowledgments	26
References	26

1. Introduction

Demands for aerospace vehicles with better performance, higher reliability and robustness, and lower cost and risk are ever increasing. To effectively address these competing objectives, designers generally take uncertainty-based methods (UBM) [1–3],

* Corresponding author.

E-mail addresses: huxingzhi@nudt.edu.cn (X. Hu), chenxiaoqian@nudt.edu.cn (X. Chen).

<http://dx.doi.org/10.1016/j.paerosci.2016.07.004>

0376-0421/© 2016 Elsevier Ltd. All rights reserved.

with consideration of a large number of uncertainties arising from the vehicle itself, as well as environmental and operational conditions. Structural design [4–6], for instance, includes model prediction errors induced by assumption and simplification and performance uncertainty caused by variations in material properties and manufacturing tolerances. These uncertainties may cause system performance to change or fluctuate, even resulting in unanticipated or even unprecedented function fault and mission failure [7,8]. It is therefore significant to fully handle these uncertainties with effective sampling-based techniques in the design optimization of aerospace vehicles [9–12].

The methodology of uncertainty-based multidisciplinary design optimization (UMDO) has received much attention due to its appealing effectiveness to enhance system reliability and robustness [13–16]. The crucial inner loop for UMDO is uncertainty quantification and propagation [17–19], i.e. uncertainty analysis. The significance and challenge of uncertainty analysis are referred to in [9,19–21], especially in multiple disciplines with complex implicit functions. The goal of uncertainty analysis differs depending on the application target. Specifically, low moments of probability density function (PDF) are often estimated for the optimization objective; the reliability level of satisfying constraints is evaluated through point-estimate of the constraint function; the complete PDF of intermediate output is computed for the input of other subsystems. To achieve all the above goals, Monte Carlo and related sampling-based methods provide an effective solution way with wide applicability [22–26].

The necessity of improved experimental and sampling-based methods is greatly emphasized in [27,28] for the UMDO of aerospace vehicles. As the model fidelity grows, more advanced Monte Carlo techniques that are able to efficiently explore the probability space of interest are required, especially in multidimensional uncertainty cases. The intention of this paper is to systematically review recent Monte Carlo based advancements in methodological and practical contributions for uncertainty-based aerospace vehicle design so that previously encountered problems can be avoided. Section 2 introduces the general employment of Monte Carlo based methods in uncertainty-based design optimization. The main focus is on the Sections 3 and 4, discussing the appealing methodological improvements and typical engineering applications, respectively. Some conclusions and recommendations for future research are presented in Section 5.

2. General Monte Carlo based methodology

Due to non-intrusive characteristics, Monte Carlo Simulation (MCS) methodology [17,29] is widely used in uncertainty-based design, analysis, and optimization problems. A common flowchart of Monte Carlo based UMDO is shown in Fig. 1. With repeated sampling and simulation, the MCS can obtain the statistics π of the output quantities of interest, which supports the whole optimization process.

As the MCS is unbiased with any level of accuracy, it can perform as a benchmark for evaluating the performance of other methods. For example, the result of uncertainty analysis is often presented through the first moment, i.e. the mean, though it is rarely enough to reflect the overall structure of π . Such a mean can be determined by

$$\pi = E[f(\mathbf{x})] = \int f(\mathbf{x})p(\mathbf{x})d\mathbf{x}, \quad (1)$$

where $f(\mathbf{x})$ can be any output function and $p(\mathbf{x})$ is the distribution function of the random vector $\mathbf{x}$.

It is worth noting that Eq. (1) can be computed with analytical methods in very rare cases since both $f(\mathbf{x})$ and $p(\mathbf{x})$ can seldom be

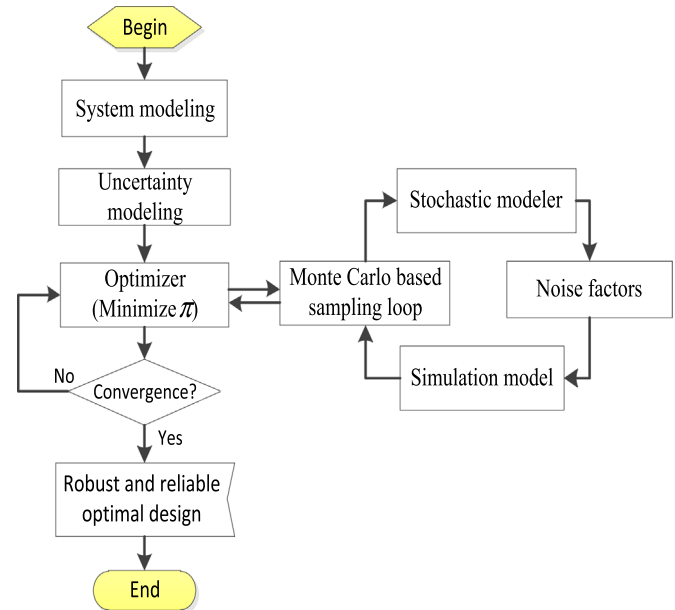


Fig. 1. General flowchart of uncertainty-based design optimization employing Monte Carlo methods.

explicitly defined, and the integration domain is also generally complicated. The simplest MCS ‘algorithm’ is drawing samples directly according to the distribution function of $\mathbf{x}$. An empirical estimate of statistics π based on perfect Monte Carlo sampling is

$$\hat{\pi} = \frac{1}{N} \sum_{i=1}^N w_i f(X^{(i)}), \quad (2)$$

where N independently distributed random samples $\{X^{(i)}; i = 1, \dots, N\}$ are generated with quadrature weights $\{w_i; i = 1, \dots, N\}$. Almost sure convergence is obtained when $N \rightarrow \infty$. Gaussian quadrature [30] and other quadrature-based or cubature-based methods [31,32] belong to this approximate approach, but they are unappealing due to the curse of dimensionality. Other approximation approaches, like the sensitivity-based method [33], e.g. first-order first-moment (FOFM), first-order second-moment (FOSM), and second-order first-moment (SOSM), employ low-order Taylor series to evaluate the integral numerically. These efforts of approximation approaches including variance reduction techniques are comprehensively given in [34].

However, the above approximate numerical integration approaches are generally only efficient and accurate for a special type of problems, e.g. quadrature-based method for polynomial response, and may not be applicable for problems with high-dimensional uncertainties and complex integrand which has no explicit formula and can only be calculated with time-consuming simulation analysis, e.g. finite element analysis (FEA) codes [35,36]. These difficulties of traditional numerical integration approaches motivate the development of Monte Carlo methods in sampling efficiency and convergence.

3. Improved Monte Carlo approaches

MCS is inherently computationally expensive, particularly when the involved deterministic calculation requires much computing time [29,37]. It can be addressed through two ways: sampling designs should be highly efficient; sampling convergence should be reliably quantified [38,39]. The latter permits halting the Monte Carlo analysis once suitably accurate outcomes have been

reached, and the former allows attaining such desired accuracy levels with a minimal number of simulation runs. To improve the both, several improved Monte Carlo methods have been recently developed and proved to be more efficient than the standard MCS.

3.1. Bayesian inference

The Bayesian inference methodology [40–42] can be employed to quantify the uncertain variables via the availability of prior information. The output statistics $\pi(\cdot)$ in this case is defined as the posterior density $p(\cdot|y)$. As for a prior belief $p(\mathbf{x})$, the posterior density $p(\mathbf{x}|y)$ is constructed as follows:

$$p(\mathbf{x}|y) = \frac{p(y|\mathbf{x})p(\mathbf{x})}{p(y)}, \quad (3)$$

where $p(\mathbf{x}|y)$ is computed through the proposition under the noisy output y [43] denoted as a likelihood function $p(y|\mathbf{x})$. The density function of $p(\mathbf{x}|y)$ can be solved numerically by Markov Chain Monte Carlo (MCMC), which continues to evolve, e.g. from random walk proposals to Hamiltonian Monte Carlo [44,45].

One of the major difficulties within the Bayesian inference is the calculation of

$$p(y) = \int p(y|\mathbf{x})p(\mathbf{x})d\mathbf{x}, \quad (4)$$

i.e. a constant which ensures the correct normalization of $p(\mathbf{x}|y)$. This constant is often complicated to solve since the integral is analytically intractable due to the high dimensionality in most practical problems, leading to the use of approximation techniques. A point estimate strategy [46–49] is often applied to address the analytically intractable issue of Eq. (4), which utilizes a suitable loss function that penalizes the probabilistic estimates, i.e. evaluating the quality of uncertainty propagation. The loss function defines the optimal Bayesian estimator by minimizing the expected loss, which is also analytically intractable. An alternative estimator is minimizing the mean square error, which has a loss function defined as $(\mathbf{x} - \hat{\mathbf{x}})^T C (\mathbf{x} - \hat{\mathbf{x}})$, where C is a positive definite matrix. The minimum mean square error (MMSE) estimate [50] is therefore given as

$$\hat{\mathbf{x}} = \int \mathbf{x} p(\mathbf{x}|y) d\mathbf{x}, \quad (5)$$

where MMSE equals to the posterior mean. Another popular estimate/analysis way is maximum-a-posterior (MAP), like $\hat{\mathbf{x}} = \arg \max p(\mathbf{x}|y)$. $\hat{\mathbf{x}}$ with the largest probability is chosen to be the MAP estimate for uncertainty analysis. These efficient point estimates eliminate the computational burdens within the practical use of Bayesian inference.

3.2. Importance sampling

In many practical situations, it is difficult to draw samples directly from the variable distributions and obtain the objective statistics π . To tackle this problem, a feasible method is to introduce a convenient distribution which enables the sampling and the density to be evaluated point by point. This improved method is known as importance sampling (IS), also referred to as weighted sampling [51,52].

IS is able to achieve uncertainty analysis for any concerned function $f(\mathbf{x})$, by drawing samples from an intermediate importance function $Q(\mathbf{x})$. Each sample is assigned an importance weight $W(\mathbf{x})$, which is equal to the Radon-Nikodym derivative [43,53] of the distribution of π with respect to the importance distribution Q . The weight is a density that integrates to one by definition [54], satisfying

$$W(\mathbf{x}) = \frac{\pi(\mathbf{x})}{Q(\mathbf{x})}, \quad (6)$$

where $\pi = d\pi/dv$ and $\tilde{Q} = dQ/dv$ denote the Radon-Nikodym derivatives of the objective output and the importance distribution, respectively. If Eq. (6) is permitted, an approximate output of interest based on N selected samples is

$$\hat{\pi}(f) = \sum_{i=1}^N W(x_i) f(x_i), \quad (7)$$

where particular choices of the importance function accelerate the convergence of the output estimate.

To determine the proper weights in Eq. (5), the definition of importance function is quite important. Particularly, one interpretation of this definition is that an unbiased approximation of the weight should be determined to the properly weighted samples and the confident convergence of π . IS is therefore pervasively studied as it is expected to reduce the biased variance to zero with carefully selected importance density function [55]. The approaches for selection of optimum IS function are discussed in [25,56–59]. Such a popular selection approach is to minimize

$$\text{var}_Q[W(\mathbf{x})] = \int \left[\frac{\pi(\mathbf{x})}{Q(\mathbf{x})} - 1 \right]^2 Q(\mathbf{x}) d\mathbf{x}, \quad (8)$$

where it ensures the $\tilde{Q}$ to be as close as possible to $\tilde{\pi}$.

Many problems in statistics and engineering require the consideration of a sequence of distributions. For example, when estimating the location/velocity of multiple objects from incoming sensor measurements in launch vehicle operations, it is more common to consider a sequence of distributions $\{\pi_t\}_{t \geq 1}$ rather than a static distribution π . Sequential importance sampling (SIS) [48,60] has been developed to allow such inference in general sequences of distributions. Compared with IS, SIS is to obtain a set of weighted random samples sequentially from $\{\pi_t\}_{t \geq 1}$, i.e. obtain samples from π_1 then π_2 and so on. These random samples are also known as particles and propagated through time. By replicating the particles proportional to their weights at each time, a resampling scheme improves the efficiency of SIS and different resampling techniques are compared in [61]. More advancements of SIS techniques can be found in [62–64].

3.3. Latin hypercube sampling

Studies have shown that the standard MCS is not an efficient sampling strategy due to its less appealing space-filling and non-collapsing characteristics [17,65–67]. Many attempts have been made during the past decades to develop more efficient sampling strategies. A widely used strategy is Latin hypercube sampling (LHS) [65,68]. As a stratified sampling approach, LHS ensures that the resulting samples are non-collapsing and generally more space-filling than random sampling [69]. It provides an estimator, i.e. the expectation of the output function, with lower variance. For N samples drawn from multiple random variables, each variable of the sample is marginally stratified. The number of strata equals the size of samples N and the probability of falling into each of the strata is $1/N$. The range of each variable is divided into N equally probable intervals, i.e. strata, and a random sample is taken at each interval. This sampling scheme makes the N samples cover all the intervals of whole variables and does not require more samples when the dimensionality of variables is increased.

There are consequently many recent examples for the use of LHS. However, the general LHS does not necessarily lead to optimal space-filling designs [39]. Several optimized Latin hypercube sampling (OLHS) strategies have therefore been studied [70,71], aiming at improving the space-filling efficacy of LHS designs.

These OLHS strategies take a basic LHS design as an initial design and then iteratively optimize the location of samples with respect to a space-filling criterion until a set of termination criteria are satisfied. Some studies have illustrated the superior efficiency of OLHS compared to MCS and LHS [72], most notably in cases where uncertainty analysis is based on a small or medium number of simulation-based samples. These OLHS strategies [73,74] mainly differ from their optimization algorithms and space-filling criteria used as the objective function. Further studies in the improvement of OLHS efficiency are also the current research emphasis in this area.

3.4. Meta-modeling approximation based MCS

Apart from the use of more efficient sampling strategies, MCS can also be accelerated through appropriate meta-modeling approximation by the data-driven and physics-free computation of the output response [75]. Sensitivity analysis or dimension reduction strategies are often incorporated to facilitate the construction of meta-models in high-dimensional cases. The cheap meta-models replace the original physical model and therefore significantly reduce the computational burden of MCS. Some meta-modeling methods currently used are response surface method (RSM), radial basis function (RBF), neural networks (NN), support vector machine (SVM), non-intrusive polynomial chaos expansions (NIPCE), Gaussian process (GP) [76–78], etc. These methods all require a training set of samples, which is selected through deterministic or random samplings. The number of model simulations required for an adequate training set is often much smaller than that required for the direct MCS.

Generating the required training set is still computationally problematic when dealing with multidisciplinary models with highly computational demand. Hence there is a kind of design of experiment (DOE) issue in the selection of the training set. The training samples should be spread over the input space so that no prediction is too far from a training point. It implies that the sampling is better to be space-filling and non-collapsing, leading to a similar issue of MCS, i.e. a limited number of available samples. This indicates the importance of sampling efficiency in the generation of the training set for the construction of meta-models, which has been a common topic within the meta-modeling community [79–81]. The improved sampling methods, like IS and OLHS, can effectively provide more efficient training sets for the meta-model construction. For example, a Kriging-based adaptive IS is developed for rare event probability estimation [82], which shows the efficiency of the approach in terms of accuracy and low number of samples.

4. Aerospace vehicle applications

The applications of improved Monte Carlo methods in aerospace vehicle design optimization are for both component-level [83–86] and system-level [87–89], in which four typical engineering applications are detailed to illustrate the use of the aforementioned approaches.

4.1. Aerospace gas turbine engine prognostics

To be more competitive, there is a growing need for gas turbine engines to minimize whole life costs, which requires the incorporation of prognostics that aims to increase the fidelity of failure-time predictions. The premise of prognostics is to estimate the remaining useful life (RUL) [90] by using sensor measurements. However, a gas turbine engine is a multidisciplinary system comprising interacted subsystems. The conventional model-based

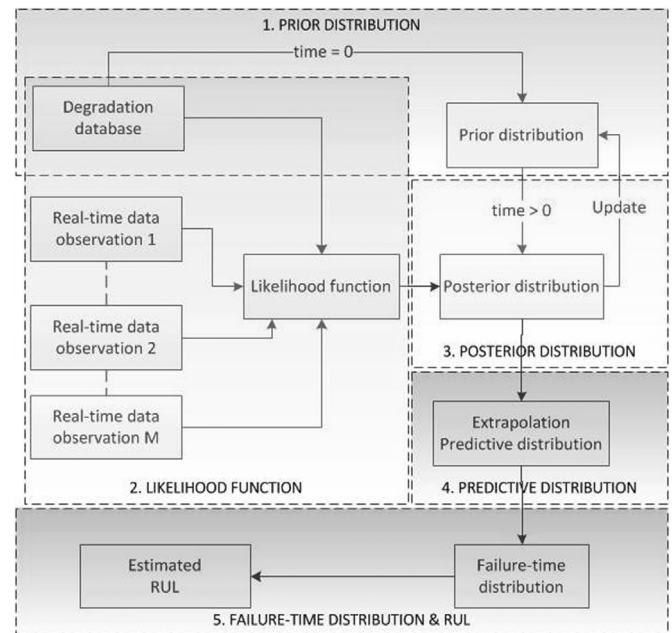


Fig. 2. Bayesian inference components for aerospace engine prognostics.

approach is difficult to achieve, thus encouraging the use of measured in-flight data with Bayesian inference that simulates the system degradation.

Data-driven Bayesian inference is desirable for this probabilistic estimation. This Bayesian inference can be divided into two kinds by prognostic models: Bayesian non-Hierarchical Model (BnHM) and Bayesian Hierarchical Model (BHM) [91,92], which belongs to a single-level model and a multi-level model, respectively. Hierarchical models are stratified and the model coefficients can vary at more than one level. A BHM is established to perform the Bayesian inference for accurate prognostics and compared with some existing BnHM in [93]. With the ability to integrate multiple kinds of data, the BHM exhibits the improvements in

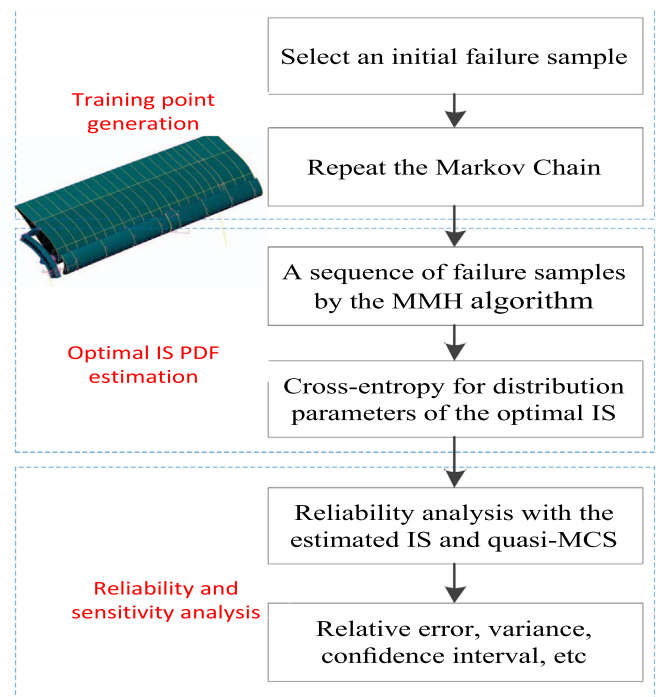


Fig. 3. Application of improved IS to reliability analysis of aircraft inside flap.

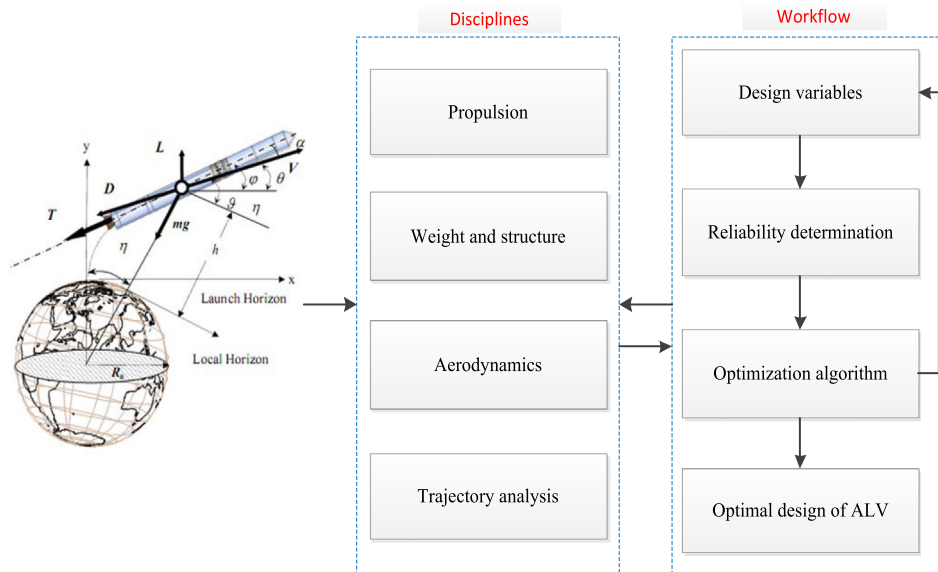


Fig. 4. Discipline modules and RBMDO workflow for aerospace launch vehicles.

accuracy and consistency to address realistic prognostic challenges, involving five main components [93], as shown in Fig. 2. Components 1–3 can be used to accommodate multiple measured data, compute the likelihood function, and estimate the posterior statistics by MCMC. The predictive distribution is inferred over a desired period in component 4 and then finally transformed into a failure time distribution. These components constitute a common application flow of data-driven Bayesian inference.

4.2. Reliability analysis of aircraft inside flap

The inside flap reliability of an aircraft is important to analyze [94,95]. As high-dimensional FEA models are considered for reliability analysis in recent years, researchers have made great efforts to develop corresponding accurate and efficient methods. However, such a non-linear problem with multiple failure modes is still a challenge. Analytical or semi-analytical approaches are not suitable for this kind of problems owing to low failure probability and high non-linearity. The standard MCS satisfies the flexibility but the computational cost is prohibitive, especially in the large-scale simulations. For example, in the FEA of a flap referred to [96], it has 2001 nodes and 3274 elements.

An improved IS method proposed for this problem is based on a cross-entropy strategy [25] and a modified Metropolis-Hastings algorithm (MMH) [97]. The workflow of this improved IS method is sketched in Fig. 3. The cross-entropy strategy is employed to estimate the optimal IS PDF with failure sample points generated according to the original PDF of the target. Instead of using expensive MCS to simulate the failure sample points, the MMH algorithm can efficiently produce such points distributed as the original PDF. After the distribution parameters of the optimal IS are obtained, the IS method is transformed into a quasi-MCS [98] for the reliability analysis. The confidence interval and its length for the failure probability are inferred simultaneously. This efficient IS method is demonstrated by several numerical examples in terms of acceptable accuracy and computational cost, along with an aircraft inside flap to show the engineering application [96].

4.3. RBMDO of aerospace launch vehicles

The design optimization of complex systems such as aerospace launch vehicles (ALVs) requires making appropriate compromises

among multiple coupled objectives, which motivates the development of reliability-based multidisciplinary design optimization (RBMDO) [99–101]. Researchers recently develop procedures to couple reliability analysis and deterministic optimization, and some successful examples of these applications can be found in [102–105]. The analysis modules and the overall workflow of RBMDO for ALVs are shown in Fig. 4.

The goal of RBMDO is to ensure the reliability under various sources of uncertainty, i.e. increase the lifetime and reduce the probability of failure [27,106]. A general reliability-based optimization problem is formulated as

$$\begin{cases} \text{find } \mu_{\mathbf{x}} \\ \text{min } \mu_f(\mathbf{x}) \\ \text{s.t. } \mathbf{R} - \mathbf{P}\{\mathbf{g}(\mathbf{x}) \leq 0\} \leq 0' \\ \mathbf{x}^l \leq \mathbf{x} \leq \mathbf{x}^u \end{cases} \quad (9)$$

where $\mathbf{x} \in (\mathbf{x}^l, \mathbf{x}^u)$ is the vector of random variables, often including design variables and other uncertain parameters. μ_f is the mean of the objective function, $\mathbf{g}$ is the constraint vector which needs to satisfy the desired reliability vector $\mathbf{R}$ with the probability statement $\mathbf{P}\{\cdot\}$.

Among the methods for determining the probability of failure or reliability, MCS is the simplest approach but too computationally expensive to achieve. As improvement, LHS is more widely used as stratified sampling to ensure the representation of all areas in the design space [107]. The sample space is divided into mutually exclusive regions and the samples from each region are taken based on the assigned distribution of each random variable. The equation for a reliability determination is

$$P\{\mathbf{g}(\mathbf{x}) \leq 0\} = \sum_{j=1}^N \left[P_j \frac{1}{M_j} \sum_{i=1}^{M_j} I_{\text{const}}(x_i) \right], \quad (10)$$

where $I_{\text{const}} = 1$ if the constraint is satisfied; otherwise, $I_{\text{const}} = 0$. P_j is the probability of j th region ($j = 1, \dots, N$) and M_j is the number of samples within the j th region. The reliability determination will be repeated until the optimization process is convergent to an optimal feasible solution.

Acknowledgments

This work was supported by National Nature Science Foundation of China [Grant nos. 91216201 and 51205403]; Fund of Innovation by Graduate School of National University of Defense Technology [Grant no. B130106]; and Fund of Innovation by Hunan Province, China [Grant no. CX2013B005]. The authors would like to thank Prof. Yong Zhao and Dr. Pranay Seshadri for their valuable comments and suggestions in improving this paper.

References

- [1] W. Yao, X.Q. Chen, W.C. Luo, M. van Tooren, J. Guo, Review of uncertainty-based multidisciplinary design optimization methods for aerospace vehicles, *Progress. Aerosp. Sci.* 47 (2011) 450–479.
- [2] L.G. Crespo, S.P. Kenny, Special edition on uncertainty quantification of the AIAA journal of aerospace computing, information, and communication, *J. Aerosp. Inf. Syst.* 12 (2015), 9–9.
- [3] N.V. Sahinidis, Optimization under uncertainty: state-of-the-art and opportunities, *Comput. Chem. Eng.* 28 (2004) 971–983.
- [4] W. Yao, J. Guo, X. Chen, M. Van Tooren, Utilizing uncertainty multi-disciplinary design optimization for conceptual design of space systems, in: *Proceedings of the 8th Annual Conference on Systems Engineering Research*, CSER 2010, Hoboken, NJ, USA, 17–19 March 2010, pp. 324–333.
- [5] W. Yao, X.Q. Chen, Y.Y. Huang, Z. Gurdal, M. van Tooren, Sequential optimization and mixed uncertainty analysis method for reliability-based optimization, *AIAA J.* 51 (2013) 2266–2277.
- [6] Alan S. Uebelhart, *Non-Deterministic Design and Analysis of Parameterized Optical Structures During Conceptual Design* (PhD dissertation), Massachusetts Institute of Technology, 2007.
- [7] Y. Miyazawa, Current status of Japanese aerospace programs—focusing on the high speed flight demonstration, *Autom. Control Aerosp.* 2004 1 (2005) 25.
- [8] D.M. Harland, R. Lorenz, *Space Systems Failures: Disasters and Rescues of Satellites, Rocket and Space Probes*, Springer Science & Business Media, New York, 2007.
- [9] L.G. Crespo, S.P. Kenny, D.P. Giesy, The NASA Langley multidisciplinary uncertainty quantification challenge, in: *Proceedings of the 16th AIAA Non-Deterministic Approaches Conference*: AIAA Reston, VA, 2014, p. 2014-1347.
- [10] S.L. Padula, C.R. Gumbert, W. Li, Aerospace applications of optimization under uncertainty, *Optim. Eng.* 7 (2006) 317–328.
- [11] C. Villa, E. Loziquez, R. Labayrade, Multi-objective optimization under uncertain objectives: application to engineering design problem, in: *Evolutionary Multi-Criterion Optimization*, Springer, Berlin, Heidelberg, 2013, pp. 796–810.
- [12] Z. Wang, W. Huang, L. Yan, Multidisciplinary design optimization approach and its application to aerospace engineering, *Chin. Sci. Bull.* 59 (2014) 5338–5353.
- [13] M. Allen, K. Maute, Reliability-based design optimization of aeroelastic structures, *Struct. Multidiscip. Optim.* 27 (2004) 228–242.
- [14] A. Messac, A. Ismail-Yahaya, Multiobjective robust design using physical programming, *Struct. Multidiscip. Optim.* 23 (2002) 357–371.
- [15] D.M. Frangopol, K. Maute, Life-cycle reliability-based optimization of civil and aerospace structures, *Comput. Struct.* 81 (2003) 397–410.
- [16] Z.P. Mourelatos, J. Liang, A methodology for trading-off performance and robustness under uncertainty, *J. Mech. Des.* 128 (2006) 856.
- [17] J.C. Helton, J.D. Johnson, C.J. Sallaberry, C.B. Storlie, Survey of sampling-based methods for uncertainty and sensitivity analysis, *Reliab. Eng. Syst. Saf.* 91 (2006) 1175–1209.
- [18] O. Schwabe, E. Shehab, J. Erkoyuncu, Uncertainty quantification metrics for whole product life cycle cost estimates in aerospace innovation, *Progress. Aerosp. Sci.* 77 (2015) 1–24.
- [19] O.P. Le Maître, O.M. Knio, *Introduction: Uncertainty Quantification and Propagation*, Springer, New York, 2010.
- [20] R.C. Smith, *Uncertainty Quantification: Theory, Implementation, and Applications*, SIAM, Philadelphia, 2013.
- [21] H. Agarwal, J.E. Renaud, E.L. Preston, D. Padmanabhan, Uncertainty quantification using evidence theory in multidisciplinary design optimization, *Reliab. Eng. Syst. Saf.* 85 (2004) 281–294.
- [22] D.P. Kroese, T. Taimre, Z.I. Botev, *Handbook of Monte Carlo Methods*, John Wiley & Sons, New York, 2013.
- [23] E. Zio, *The Monte Carlo Simulation Method for System Reliability and Risk Analysis*, Springer, New York, 2013.
- [24] A. Smith, A. Doucet, N. de Freitas, N. Gordon, *Sequential Monte Carlo methods in practice*, Springer Science & Business Media, New York, 2013.
- [25] R.Y. Rubinstein, D.P. Kroese, The cross-entropy method: a unified approach to combinatorial optimization, Monte-Carlo simulation and machine learning, *Springer Science & Business Media*, New York, 2013.
- [26] N. Van Nguyen, D. Lee, H.-U. Park, J.-W. Lee, A multidisciplinary robust optimization framework for UAV conceptual design, *Aeronaut. J.* 118 (2014) 123–142.
- [27] T.A. Zang, M.J. Hemsch, M.W. Hilburger, S.P. Kenny, J.M. Luckring, P. Maghami, et al., Needs and Opportunities for Uncertainty-Based Multidisciplinary Design Methods for Aerospace Vehicles, National Aeronautics and Space Administration, Langley Research Center, Hampton, 2002.
- [28] H.-U. Park, J.-W. Lee, J. Chung, K. Behdinan, M.D.O. Uncertainty-based, for aircraft conceptual design, *Aircr. Eng. Aerosp. Technol.: Int. J.* 87 (2015) 345–356.
- [29] D.P. Landau, K. Binder, *A Guide to Monte Carlo Simulations in Statistical Physics*, Cambridge university press, Cambridge, 2014.
- [30] P.J. Davis, P. Rabinowitz, *Methods of Numerical Integration*, Courier Corporation, North Chelmsford, 2007.
- [31] K.S. Li, P. Lumb, Reliability analysis by numerical integration and curve fitting, *Struct. Saf.* 3 (1985) 29–36.
- [32] Y. Lin, *A Quadrature-Based Technique for Robust Design With Computer Simulations* (PhD dissertation), Massachusetts Institute of Technology, 2007.
- [33] H. Rue, S. Martino, N. Chopin, Approximate Bayesian inference for latent Gaussian models by using integrated nested Laplace approximations, *J. R. Stat. Soc.: Ser. B (Stat. Methodol.)* 71 (2009) 319–392.
- [34] J.F. Monahan, *Numerical Methods of Statistics*, Cambridge University Press, Cambridge, 2011.
- [35] T. Kitamura, K. Yoshida, H. Seko, Q. Shi, Analysis of shock response for satellite separation, in: *Proceedings of International Conference on Noise and Vibration Engineering (Isma2012)/International Conference on Uncertainty In Structural Dynamics (Usd2012)*, 2012, pp. 2091–2101.
- [36] E. Vanmarcke, M. Grigoriu, Stochastic finite element analysis of simple beams, *J. Eng. Mech.* 109 (1983) 1203–1214.
- [37] C. Xu, H.S. He, Y. Hu, Y. Chang, X. Li, R. Bu, Latin hypercube sampling and geostatistical modeling of spatial uncertainty in a spatially explicit forest landscape model simulation, *Ecol. Model.* 185 (2005) 255–269.
- [38] V.R. Joseph, Y. Hung, Orthogonal-maximin Latin hypercube designs, *Stat. Sin.* (2008) 171–186.
- [39] L. Pronzato, W.G. Müller, Design of computer experiments: space filling and beyond, *Stat. Comput.* 22 (2012) 681–701.
- [40] P.M. Lee, *Bayesian Statistics: An Introduction*, John Wiley & Sons, New York, 2012.
- [41] S.J. Press, J.S. Press, *Bayesian Statistics: Principles, Models, and Applications*, Wiley, New York, 1989.
- [42] J.M. Bernardo, A.F. Smith, *Bayesian Theory*, IOP Publishing, 2001.
- [43] M. Briers, *Improved Monte Carlo Methods for State-Space Models* (PhD dissertation), University of Cambridge, 2007.
- [44] S. Sankaranarayanan, Uncertainty reduction using Bayesian inference and sensitivity analysis: a sequential approach to the NASA Langley uncertainty quantification challenge, in: *Proceedings of the 18th AIAA Non-Deterministic Approaches Conference*, 2016, p. 1194.
- [45] T. Hastie, R. Tibshirani, J. Friedman, J. Franklin, The elements of statistical learning: data mining, inference and prediction, *Math. Intell.* 27 (2005) 83–85.
- [46] P.J. Huber, *Robust Statistics*, Springer, New York, 2011.
- [47] A.D. Martin, K.M. Quinn, Dynamic ideal point estimation via Markov chain Monte Carlo for the US Supreme Court, 1953–1999, *Political Anal.* 10 (2002) 134–153.
- [48] A. Doucet, S. Godsill, C. Andrieu, On sequential Monte Carlo sampling methods for Bayesian filtering, *Stat. Comput.* 10 (2000) 197–208.
- [49] O. Cappé, S.J. Godsill, E. Moulines, An overview of existing methods and recent advances in sequential Monte Carlo, *Proc. IEEE* 95 (2007) 899–924.
- [50] B. Efron, C.N. Morris, Stein's Paradox in Statistics, *Sci. Am.* 236 (1977) 119–127.
- [51] C. Robert, G. Casella, *Monte Carlo Statistical Methods*, Springer Science & Business Media, New York, 2013.
- [52] J. Martin, L.C. Wilcox, C. Burstedde, O. Ghattas, A stochastic Newton MCMC method for large-scale statistical inverse problems with application to seismic inversion, *SIAM J. Sci. Comput.* 34 (2012) A1460–A1487.
- [53] A.W. Marshall, The Use of Multistage Sampling Schemes in Monte Carlo Computations, 1954.
- [54] A. Jasra, *Bayesian Inference for Mixture Models via Monte Carlo Computation* (PhD dissertation), Imperial College London (University of London), 2006.
- [55] G.L. Ang, A.H.-S. Ang, W.H. Tang, Optimal importance-sampling density estimator, *J. Eng. Mech.* 118 (1992) 1146–1163.
- [56] Y. Chen, Another look at rejection sampling through importance sampling, *Stat. Probab. Lett.* 72 (2005) 277–283.
- [57] A. Hinrichs, Optimal importance sampling for the approximation of integrals, *J. Complex.* 26 (2010) 125–134.
- [58] G. Jiang, C. Xu, M.C. Fu, On sample average approximation algorithms for determining the optimal importance sampling parameters in pricing financial derivatives on Lévy processes, *Oper. Res. Lett.* 44 (2016) 44–49.
- [59] D.P. Kroese, R.Y. Rubinstein, I. Cohen, S. Porotsky, T. Taimre, Cross-entropy method, in: *Encyclopedia of Operations Research and Management Science*, Springer, US, 2013, pp. 326–333.
- [60] S.N. MacEachern, M. Clyde, J.S. Liu, Sequential importance sampling for nonparametric Bayes models: the next generation, *Can. J. Stat.* 27 (1999) 251–267.
- [61] R. Douc, O. Cappé, Comparison of resampling schemes for particle filtering, in: *ISPA 2005 Proceedings of the 4th International Symposium on Image and Signal Processing and Analysis*, IEEE, 2005, pp. 64–69.
- [62] A. Doucet, M. Briers, S. Sénécal, Efficient block sampling strategies for sequential Monte Carlo methods, *J. Comput. Graph. Stat.* (2012).
- [63] N. Whiteley, A. Lee, K. Heine, On the role of interaction in sequential Monte

- Carlo algorithms, *Bernoulli* 22 (2016) 494–529.
- [64] Y. Wu, J. Wang, X. Lu, Y. Cao, Modified sequential importance resampling filter, *J. Syst. Eng. Electron.* 26 (2015) 441–449.
- [65] J.C. Helton, F.J. Davis, Latin hypercube sampling and the propagation of uncertainty in analyses of complex systems, *Reliab. Eng. Syst. Saf.* 81 (2003) 23–69.
- [66] B.G. Husslage, G. Rennen, E.R. van Dam, D. den Hertog, Space-filling Latin hypercube designs for computer experiments, *Optim. Eng.* 12 (2011) 611–630.
- [67] H. Janssen, Monte-Carlo based uncertainty analysis: Sampling efficiency and sampling convergence, *Reliab. Eng. Syst. Saf.* 109 (2013) 123–132.
- [68] A.B. Owen, Controlling correlations in Latin hypercube samples, *J. Am. Stat. Assoc.* 89 (1994) 1517–1522.
- [69] M. Vořechovský, D. Novák, Statistical correlation in stratified sampling, in: *Proceedings of the 9th Int Conf on Applications of Statistics and Probability in Civil Engineering—ICASP, CiteSeer*, 2003, pp. 119–124.
- [70] M. Liefvendahl, R. Stocki, A study on algorithms for optimization of Latin hypercubes, *J. Stat. Plan. Inference* 136 (2006) 3231–3247.
- [71] T.M. Cioppa, T.W. Lucas, Efficient nearly orthogonal and space-filling Latin hypercubes, *Technometrics* (2012).
- [72] M.M. Rajabi, B. Ataie-Ashtiani, H. Janssen, Efficiency enhancement of optimized Latin hypercube sampling strategies: application to Monte Carlo uncertainty analysis and meta-modeling, *Adv. Water Resour.* 76 (2015) 127–139.
- [73] R. Stocki, A method to improve design reliability using optimal Latin hypercube sampling, *Comput. Assist. Mech. Eng. Sci.* 12 (2005) 393.
- [74] Q.Y. Kenny, W. Li, A. Sudjianto, Algorithmic construction of optimal symmetric Latin hypercube designs, *J. Stat. Plan. Inference* 90 (2000) 145–159.
- [75] A. Forrester, A. Sobester, A. Keane, *Engineering Design via Surrogate Modelling: a Practical Guide*, John Wiley & Sons, New York, 2008.
- [76] R. Jin, W. Chen, T.W. Simpson, Comparative studies of metamodelling techniques under multiple modelling criteria, *Struct. Multidiscip. Optim.* 23 (2001) 1–13.
- [77] T.T. Allen, M.A. Bernshteyn, K. Kabiri-Bamradian, Constructing meta-models for computer experiments, *J. Qual. Technol.* 35 (2003) 264.
- [78] S. Razavi, B.A. Tolson, D.H. Burn, Review of surrogate modeling in water resources, *Water Resour. Res.* 48 (2012).
- [79] R.T. Johnson, D.C. Montgomery, B. Jones, P.A. Parker, Comparing computer experiments for fitting high-order polynomial metamodels, *J. Qual. Technol.* 42 (2010) 86.
- [80] F.M. Alam, K.R. McNaught, T.J. Ringrose, A comparison of experimental designs in the development of a neural network simulation metamodel, *Simul. Model. Pract. Theory* 12 (2004) 559–578.
- [81] F. Boukouvala, M.G. Ierapetritou, Feasibility analysis of black-box processes using an adaptive sampling Kriging-based method, *Comput. Chem. Eng.* 36 (2012) 358–368.
- [82] M. Balesdent, J. Morio, J. Marzat, Kriging-based adaptive importance sampling algorithms for rare event estimation, *Struct. Saf.* 44 (2013) 1–10.
- [83] M. Nikbay, P. Acar, A.R. Aslan, Reliability based design optimization of a CubeSat de-orbiting mechanism, in: *Proceedings of the 12th AIAA Aviation Technology, Integration, and Operations (ATIO) Conference and 14th AIAA/ISSMO Multidisciplinary Analysis and Optimization Conference*, p. 5711.
- [84] M. Pellissetti, E. Capiez-Lernout, H. Pradlwarter, C. Soize, G. Schueller, Reliability analysis of a satellite structure with a parametric and a non-parametric probabilistic model, *Comput. Methods Appl. Mech. Eng.* 198 (2008) 344–357.
- [85] A. Manan, J. Cooper, Design of composite wings including uncertainties: a probabilistic approach, *J. Aircr.* 46 (2009) 601–607.
- [86] H. Fang, H. Shi, Y. Xiong, R. Li, P. Wang, The component-level and system-level satellite power system health state evaluation method, in: *Proceedings of the Prognostics and System Health Management Conference (PHM-2014 Hunan)*, IEEE, 2014, pp. 683–688.
- [87] L. Brévault, M. Balesdent, N. Bérend, R. Le Riche, Challenges and future trends in uncertainty-based multidisciplinary design optimization for space transportation system design, in: *Proceedings of the 5th European Conference for Aeronautics and Space Sciences (EUCASS 2013)*, 2013.
- [88] D.A. Delaurentis, A Probabilistic Approach to Aircraft Design Emphasizing Guidance and Stability and Control Uncertainties, Georgia Institute of Technology, Atlanta, 1998.
- [89] J.T. Hwang, D.Y. Lee, J.W. Cutler, J.R. Martins, Large-scale multidisciplinary optimization of a small satellite's design and operation, *J. Spacecr. Rockets* 51 (2014) 1648–1663.
- [90] S. Qiang, J.G. Yue, Review on fault prognostic methods based on uncertainty, Kongzhi Yu Juece/Control Decis. 29 (2014) 769–778.
- [91] L. Zhuang, *Bayesian Dynamical Modeling of Count Data* (PhD dissertation), The Ohio State University, 2011.
- [92] A. Gelman, Prior distributions for variance parameters in hierarchical models (comment on article by Browne and Draper), *Bayesian Anal.* 1 (2006) 515–534.
- [93] M.A. Zaidan, R.F. Harrison, A.R. Mills, P.J. Fleming, Bayesian hierarchical models for aerospace gas turbine engine prognostics, *Expert Syst. Appl.* 42 (2015) 539–553.
- [94] Z. Tang, Z. Lu, D. Li, Z. Feng, Nonprobabilistic reliability analysis for an inside flap of an aircraft, *J. Aircr.* 49 (2012) 250–256.
- [95] J.W. Bennett, B.C. Mecrow, A.G. Jack, D.J. Atkinson, A prototype electrical actuator for aircraft flaps, *Ind. Appl. IEEE Trans.* 46 (2010) 915–921.
- [96] Z. Tang, Z. Lu, J. Feng, B. Wang, The applications of an importance sampling method to reliability analysis of the inside flap of an aircraft, *Proc. Inst. Mech. Eng. Part G: J. Aerosp. Eng.* 227 (2013) 916–932.
- [97] S. Chib, E. Greenberg, Understanding the metropolis-hastings algorithm, *Am. Stat.* 49 (1995) 327–335.
- [98] H. Niederreiter, *Quasi-Monte Carlo Methods*, John Wiley & Sons, New York, 2010.
- [99] R.M. Paiva, C. Crawford, A. Suleman, Robust and reliability-based design optimization framework for wing design, *AIAA J.* 52 (2014) 711–724.
- [100] J. Roshanian, M. Ebrahimi, Latin hypercube sampling applied to reliability-based multidisciplinary design optimization of a launch vehicle, *Aerosp. Sci. Technol.* 28 (2013) 297–304.
- [101] B.D. Youn, K.K. Choi, Selecting probabilistic approaches for reliability-based design optimization, *AIAA J.* 42 (2004) 124–131.
- [102] A. Akhtar, H. Linshu, An efficient evolutionary multi-objective approach for robust design of multi-stage space launch vehicle, in: *Proceedings of the 11th AIAA/ISSMO Multidisciplinary Analysis and Optimization Conference*, Portsmouth, Virginia, 2006.
- [103] M. Ebrahimi, M.R. Farmani, J. Roshanian, Multidisciplinary design of a small satellite launch vehicle using particle swarm optimization, *Struct. Multidiscip. Optim.* 44 (2011) 773–784.
- [104] M. Ebrahimi, J. Roshanian, F. Barzinpour, Multidisciplinary design optimization approach for a small solid propellant launch vehicle conceptual design using hybrid simulated annealing, in: *Applied Mechanics and Materials*, Trans Tech Publ, 2012, pp. 4765–4771.
- [105] A. Rafique, H. Linshu, Q. Zeeshan, K. Nisar, W. Xiaowei, Integrated system design of air launched small space launch vehicle using genetic algorithm, in: *Proceedings of the 45th AIAA/ASME/SAE/ASEE Joint Propulsion Conference AIAA-2009-55062009*.
- [106] A. Keane, P. Nair, *Computational Approaches for Aerospace Design: The Pursuit of Excellence*, John Wiley & Sons, New York, 2005.
- [107] J.M. Zentner, *A Design Space Exploration Process for Large Scale, Multi-Objective Computer Simulations*, 2006.
- [108] C. Liu, *Uncertainty Modeling With Clouds in Satellite System Design Optimization* (PhD dissertation), Harbin Institute of Technology, 2011.
- [109] W.R. Wu, H. Huang, S.Y. Chen, B.B. Wut, Satellite multidisciplinary design optimization with a high-fidelity model, *J. Spacecr. Rockets* 50 (2013) 463–466.
- [110] W. Edmonson, J. Chenou, N. Neogi, H. Herencia-Zapana, Small satellite systems design methodology: a formal and agile design process, in: *Proceedings of the 2014 8th Annual IEEE Systems Conference (SysCon)*, Ottawa, Ontario, IEEE, 2014, pp. 518–524.
- [111] R.W. James, J.L. Wiley, *Space Mission Analysis and Design*, Space Technology Library, 1999.
- [112] A. Jafarsalehi, H. Fazeley, M. Mirshams, Spacecraft Mission Design Optimization Under Uncertainty, *Proc. Inst. Mech. Eng. Part C: J. Mech. Eng. Sci.* (2015), 0954406215603416.
- [113] X. Hu, G.T. Parks, X. Chen, P. Seshadri, Discovering a one-dimensional active subspace to quantify multidisciplinary uncertainty in satellite system design, *Adv. Space Res.* (2016) 1268–1279.
- [114] P.G. Constantine, *Active Subspaces: Emerging Ideas for Dimension Reduction in Parameter Studies*, SIAM, Philadelphia, 2015.
- [115] X. Hu, X. Chen, V. Lattarulo, G.T. Parks, Multidisciplinary optimization under high-dimensional uncertainty for small satellite system design, *AIAA J.* 54 (2016) 1732–1741.
- [116] T.M. Russi, *Uncertainty Quantification With Experimental Data and Complex System Models*, University of California, Berkeley, 2010.
- [117] B. Efron, R.J. Tibshirani, *An Introduction to the Bootstrap*, CRC press, New York, 1994.